

CARBON AND ITS COMPOUNDS

Student Learning Pack

1. Metadata

Topic	Carbon and its Compounds
Level	Class X
Lesson Length	60 minutes

2. Learning Objectives

- LO1: Explain how carbon's tetravalency and catenation enable the formation of a vast number of compounds.
- LO2: Distinguish saturated from unsaturated carbon compounds based on the type of C-C bonds present.
- LO3: Classify hydrocarbons into alkanes, alkenes, and alkynes using bond criteria.
- LO4: Describe the historical significance of Wöhler's synthesis of urea for organic chemistry.
- LO5: Summarize typical physical properties of carbon compounds and explain their molecular basis.

3. Prerequisite Knowledge

- Understanding of covalent bonding and valence electrons.
- Ability to read and write chemical formulas (e.g., C_2H_6).
- Familiarity with single, double, and triple bonds.

4. Key Terms

Term	Definition
Tetravalency	The ability of carbon to form four covalent bonds with other atoms.
Catenation	The capability of carbon atoms to bond to each other, forming chains, branches, or rings.
Saturated compound	A carbon compound that contains only single C-C bonds.
Unsaturated compound	A carbon compound that contains one or more double or triple C-C bonds.
Hydrocarbon	A compound composed solely of carbon and hydrogen atoms.
Alkane	A saturated hydrocarbon containing only single C-C bonds.
Alkene	An unsaturated hydrocarbon containing at least one $C=C$ double bond.
Alkyne	An unsaturated hydrocarbon containing at least one $C\equiv C$ triple bond.
Benzene	A cyclic hydrocarbon with formula C_6H_6 , featuring a ring of six carbon atoms with alternating double bonds.

5. Key Points

- Carbon can form four covalent bonds and link to other carbon atoms, creating countless compounds.

- Hydrocarbons are grouped as alkanes, alkenes, or alkynes based on the type of carbon-carbon bonds they contain.
- Saturated compounds contain only single bonds; unsaturated compounds contain double or triple bonds.
- Wöhler's 1828 preparation of urea from ammonium cyanate disproved the vital-force theory of organic synthesis.
- Carbon compounds generally have low melting/boiling points and do not conduct electricity.
- Carbon atoms can arrange into straight chains, branched chains, or cyclic (ring) structures, each being saturated or unsaturated.

6. Real-World Applications

Urea Production in Agriculture

Urea, first synthesized by Wöhler, is now manufactured on an industrial scale and used as a nitrogen fertilizer for crops.

Think about: Why is the laboratory synthesis of urea historically important for modern agriculture?

Suggested time: 5 minutes

Choosing Materials Based on Electrical Conductivity

Because carbon compounds are poor conductors, they are used as insulating materials in electrical applications.

Think about: How do the non-conductive properties of carbon compounds influence their use in everyday items?

Suggested time: 4 minutes

7. Core Notes

Carbon's Tetravalency and Catenation

Carbon has an atomic number of 6 and four valence electrons, giving it a tetravalent nature. Because each carbon can bond to up to four other atoms, it can also bond to other carbon atoms. This ability, called catenation, allows the formation of long chains, branched structures, and rings. The combination of tetravalency and catenation is the reason for the very large number of carbon-containing compounds.

- Tetravalent: forms four covalent bonds
- Catenation: carbon-carbon bonding creates chains, branches, cycles

Classification of Hydrocarbons

All compounds that contain only carbon and hydrogen are called hydrocarbons. When every carbon-carbon bond is a single bond, the hydrocarbon is saturated and is named an alkane. If one or more carbon-carbon double bonds are present, the compound is unsaturated and called an alkene. When one or more carbon-carbon triple bonds occur, the compound is an alkyne. These names reflect the bond type present in the molecule.

- Alkane - only single C-C bonds (saturated)
- Alkene - at least one C=C double bond (unsaturated)
- Alkyne - at least one C≡C triple bond (unsaturated)

Saturated vs. Unsaturated Carbon Compounds

A saturated carbon compound has the maximum number of hydrogen atoms attached to each carbon because all carbon-carbon connections are single bonds. Unsaturated compounds have fewer hydrogen atoms because double or triple bonds replace some C-H bonds. For example, ethane (C_2H_6) is saturated, while ethene (C_2H_4) is unsaturated due to a C=C double bond.

- Saturated → all single C-C bonds
- Unsaturated → one or more C=C or C≡C bonds

Historical Milestone - Wöhler's Synthesis of Urea

It was once believed that organic compounds could only be produced by living organisms because a 'vital force' was required. In 1828, Friedrich Wöhler synthesized urea, an organic compound, from the inorganic ammonium cyanate. This experiment showed that organic molecules can be created in the laboratory without any vital force, marking the birth of modern organic chemistry.

Physical Properties of Carbon Compounds

Data on carbon compounds show they melt and boil at lower temperatures than ionic substances, indicating weak intermolecular forces. They are also poor conductors of electricity, which means their bonding does not produce ions. These properties arise because the molecules are held together mainly by weak van der Waals forces rather than strong ionic bonds.

- Low melting & boiling points
- Non-conductive (no ions)
- Weak intermolecular attractions

Structural Variety - Chains, Branches, and Rings

Because of catenation, carbon atoms can link in several ways:

- Straight (linear) chains - e.g., methane, ethane
 - Branched chains - carbon atoms form side-branches off the main chain
 - Cyclic structures - carbon atoms close to form a ring, such as benzene (C_6H_6)
- Each of these frameworks may be saturated (only single bonds) or unsaturated (containing double or triple bonds).

8. Activities

Bond-Type Sorting Card Game (12 min)

Students receive cards showing molecular formulas and must place each card under the correct heading: Alkane, Alkene, or Alkyne, based on the presence of single, double, or triple bonds.

Historical Role-Play: Wöhler's Experiment (10 min)

In pairs, one student acts as Friedrich Wöhler and the other as a skeptical chemist of the era. They reenact the preparation of urea from ammonium cyanate and discuss its impact on the belief in vital force.

9. Questions

Q1. What two structural features of carbon allow it to form a large variety of compounds?

Q2. Define a saturated carbon compound and give an example.

Q3. How does an unsaturated compound differ in bonding, and name two types of unsaturated hydrocarbons.

Q4. Which historical experiment disproved the 'vital force' theory, and what was synthesized?

Q5. List two typical physical properties of carbon compounds and explain why they occur.